

```

library(tidyverse)
library(ellipse)
library(mvnTest)

X <- matrix(c(2, 8, 6, 8, 12, 9, 9, 10),
            ncol = 2, byrow = FALSE)

mu_0 <- c(7, 11)

n <- nrow(X)
p <- ncol(X)
x_bar <- colMeans(X)
S <- cov(X)

# a
T2 <- n * t(x_bar - mu_0) %*% solve(S) %*% (x_bar - mu_0)
T2 <- as.numeric(T2)
cat("T-squared value:", T2, "\n")

# (b)
F_stat <- ((n - p) / (p * (n - 1))) * T2
cat("\nDistribution: F-statistic =", round(F_stat, 4),
    "with df1 =", p, "and df2 =", n-p, "\n")

# c
alpha <- 0.05
F_crit <- qf(1 - alpha, df1 = p, df2 = n - p)
T2_crit <- ((n - 1) * p / (n - p)) * F_crit

cat("\nCritical T-squared value:", T2_crit, "\n")
p_value <- 1 - pf((n - p) / ((n - 1) * p) * T2, df1 = p, df2 = n - p)
cat("P-value:", p_value, "\n")

if(T2 > T2_crit) {
  print("Reject the null hypothesis H0.\n")
} else {
  print("Fail to reject the null hypothesis H0.\n")
}

T-squared value: 13.63636

Distribution: F-statistic = 4.5455 with df1 = 2 and df2 = 2

Critical T-squared value: 57
P-value: 0.1803279
[1] "Fail to reject the null hypothesis H0.\n"

```

c. The critical T^2 value at the $\alpha=0.05$ significance level is 57. Because the calculated T^2 statistic (13.63636) is less than the critical value (57), the resulting p-value is 0.1803279. Since the p-

value is greater than 0.05, the statistical conclusion is that we fail to reject the null hypothesis H_0 .

```
n <- 61
p <- 6
alpha <- 0.05
x_bar <- c(95.52, 164.38, 55.69, 93.39, 17.98, 31.13)
names(x_bar) <- c("Weight", "Body_Length", "Neck_Girth", "Girth",
"Head_Length", "Head_Width")

S <- matrix(c(
  3266.46, 1343.97, 731.54, 1175.50, 162.68, 238.37,
  1343.97, 721.91, 324.25, 537.35, 80.17, 117.73,
  731.54, 324.25, 179.28, 281.17, 39.15, 56.80,
  1175.50, 537.35, 281.17, 474.98, 63.73, 94.85,
  162.68, 80.17, 39.15, 63.73, 9.95, 13.88,
  238.37, 117.73, 56.80, 94.85, 13.88, 21.26
), nrow = 6, byrow = TRUE)

se <- sqrt(diag(S) / n)

# a
chi2_val <- qchisq(1 - alpha, p)
sci_lower <- x_bar - sqrt(chi2_val) * se
sci_upper <- x_bar + sqrt(chi2_val) * se
sci_df <- data.frame( Lower_SCI = sci_lower, Upper_SCI = sci_upper)
cat("95% Simultaneous Confidence Intervals\n")
print(sci_df)

# c
m <- 6
t_val_bonf <- qt(1 - alpha / (2 * m), n - 1)
bonf_lower <- x_bar - t_val_bonf * se
bonf_upper <- x_bar + t_val_bonf * se
bonf_df <- data.frame(Lower_Bonf = bonf_lower, Upper_Bonf =
bonf_upper)
cat("\n95% Bonferroni Confidence Intervals \n")
print(bonf_df)

sub_x_bar <- x_bar[c(1, 4)]
sub_S <- S[c(1, 4), c(1, 4)]

radius_simult <- sqrt(qchisq(1 - alpha, p) / n)
ell_points <- ellipse::ellipse(x = sub_S, centre =
as.numeric(sub_x_bar), t = radius_simult)

plot(ell_points, type = "l", col = "blue", lwd = 2,
      xlab = "Mean Weight (kg)", ylab = "Mean Girth (cm)",
      main = "95% Simultaneous Ellipse vs Bonferroni Rectangle",
      xlim = c(sci_df$Lower_SCI[1] - 5, sci_df$Upper_SCI[1] + 5),
```

```

ylim = c(sci_df$Lower_SCI[4] - 2, sci_df$Upper_SCI[4] + 2))
rect(bonf_df$Lower_Bonf[1], bonf_df$Lower_Bonf[4],
     bonf_df$Upper_Bonf[1], bonf_df$Upper_Bonf[4],
     border = "red", lwd = 2, lty = 2)
legend("topleft", legend = c("Simultaneous Ellipse", "Bonferroni
Rectangle"),
      col = c("blue", "red"), lty = c(1, 2), lwd = 2)

```

```

# e
m_new <- 7
t_val_bonf_new <- qt(1 - alpha / (2 * m_new), n - 1)
diff_mean <- x_bar[6] - x_bar[5]

diff_se <- sqrt((S[6,6] + S[5,5] - 2 * S[5,6]) / n)
lower_diff <- diff_mean - t_val_bonf_new * diff_se
upper_diff <- diff_mean + t_val_bonf_new * diff_se
cat("\n95% Bonferroni Confidence Interval for (Head Width - Head
Length)\n")
cat(sprintf("Estimate: %.2f | CI: [%.2f, %.2f]\n", diff_mean,
lower_diff, upper_diff))

```

95% Simultaneous Confidence Intervals

	Lower_SCI	Upper_SCI
Weight	69.55347	121.48653
Body_Length	152.17278	176.58722
Neck_Girth	49.60667	61.77333
Girth	83.48823	103.29177
Head_Length	16.54687	19.41313
Head_Width	29.03513	33.22487

95% Bonferroni Confidence Intervals

	Lower_Bonf	Upper_Bonf
Weight	75.55331	115.48669
Body_Length	154.99339	173.76661
Neck_Girth	51.01229	60.36771
Girth	85.77614	101.00386
Head_Length	16.87801	19.08199
Head_Width	29.51917	32.74083

95% Bonferroni Confidence Interval for (Head Width - Head Length)
Estimate: 13.15 | CI: [12.49, 13.81]



The Simultaneous Ellipse (solid blue line) is wider and tilted because weight and girth have a positive relationship. Meanwhile, the Bonferroni Rectangle (dashed red line) is aligned with the axes and gives individual intervals. Overall, the ellipse is larger because it considers the relationship between the two variables, while the rectangle mostly stays within the ellipse but extends slightly at the corners.

```
Cr <- c(0.48, 40.53, 2.19, 0.55, 0.74, 0.66, 0.93, 0.37, 0.22)
St <- c(12.57, 73.68, 11.13, 20.03, 20.29, 0.78, 4.64, 0.43, 1.08)
data_hairs <- data.frame(Cr, St)
n <- nrow(data_hairs)
p <- 2
alpha <- 0.10

x_bar <- colMeans(data_hairs)
S <- cov(data_hairs)

# a
radius_T2 <- sqrt(((n - 1) * p / (n - p)) * qf(1 - alpha, p, n - p) /
n)
ell_points <- ellipse::ellipse(x = S, centre = as.numeric(x_bar), t =
radius_T2)

x_limits <- range(c(ell_points[, 1], data_hairs$Cr, 0.30))
y_limits <- range(c(ell_points[, 2], data_hairs$St, 10))
```

```

plot(ell_points, type = "l", col = "blue", lwd = 2,
     xlim = x_limits, ylim = y_limits,
     xlab = "Chromium (Cr)", ylab = "Strontium (St)",
     main = "90% Joint Confidence Ellipse with Data Overlay")

points(data_hairs$Cr, data_hairs$St, pch = 16, col = "gray40")

points(x_bar[1], x_bar[2], pch = 19, col = "black", cex = 1.2)

points(0.30, 10, pch = 4, col = "red", lwd = 2, cex = 1.5)
text(0.30, 10, "[0.30, 10]", pos = 4, col = "red")

# b
se <- sqrt(diag(S) / n)
multiplier <- sqrt(((n - 1) * p / (n - p)) * qf(1 - alpha, p, n - p))
lower_ci <- x_bar - multiplier * se
upper_ci <- x_bar + multiplier * se
sci_df <- data.frame(Variable = c("Cr", "St"), Lower = lower_ci, Upper
= upper_ci)
cat("90% Simultaneous Confidence Intervals\n")
print(sci_df)

mu_test <- c(0.30, 10)
T2_test <- n * t(x_bar - mu_test) %*% solve(S) %*% (x_bar - mu_test)
T2_crit <- ((n - 1) * p / (n - p)) * qf(1 - alpha, p, n - p)
cat("\nT^2 statistic for [0.30, 10]:", as.numeric(T2_test), "\n")
cat("Critical Value:", T2_crit, "\n")

# c
par(mfrow = c(1, 3))
qqnorm(data_hairs$Cr, main = "Q-Q Plot: Chromium");
qqline(data_hairs$Cr)
qqnorm(data_hairs$St, main = "Q-Q Plot: Strontium");
qqline(data_hairs$St)
plot(data_hairs$Cr, data_hairs$St, main = "Scatter Diagram", xlab =
"Cr", ylab = "St")
par(mfrow = c(1, 1))

# d
outlier_idx <- which.max(data_hairs$Cr)
data_clean <- data_hairs[-outlier_idx, ]
n_clean <- nrow(data_clean)
x_bar_clean <- colMeans(data_clean)
S_clean <- cov(data_clean)

T2_test_clean <- n_clean * t(x_bar_clean - mu_test) %*% solve(S_clean)
%*% (x_bar_clean - mu_test)
T2_crit_clean <- ((n_clean - 1) * p / (n_clean - p)) * qf(1 - alpha,
p, n_clean - p)

```

```
cat("\nAfter Outlier Removal\n")
cat("T^2 statistic for [0.30, 10]:", as.numeric(T2_test_clean), "\n")
cat("Critical Value:", T2_crit_clean, "\n")
```

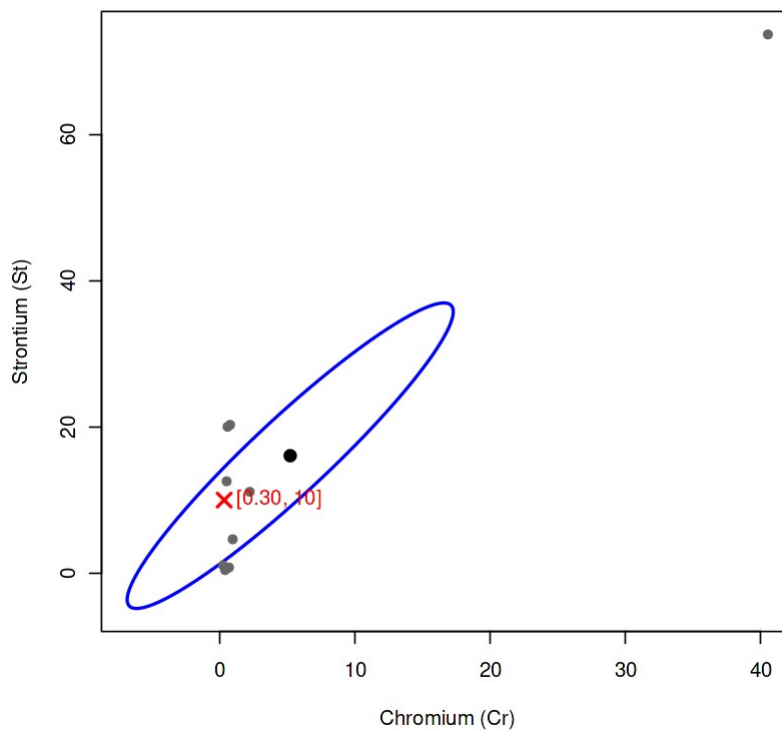
90% Simultaneous Confidence Intervals

Variable	Lower	Upper
Cr	Cr -6.881173	17.25228
St	St -4.826957	36.96696

T^2 statistic for [0.30, 10]: 1.772481

Critical Value: 7.445582

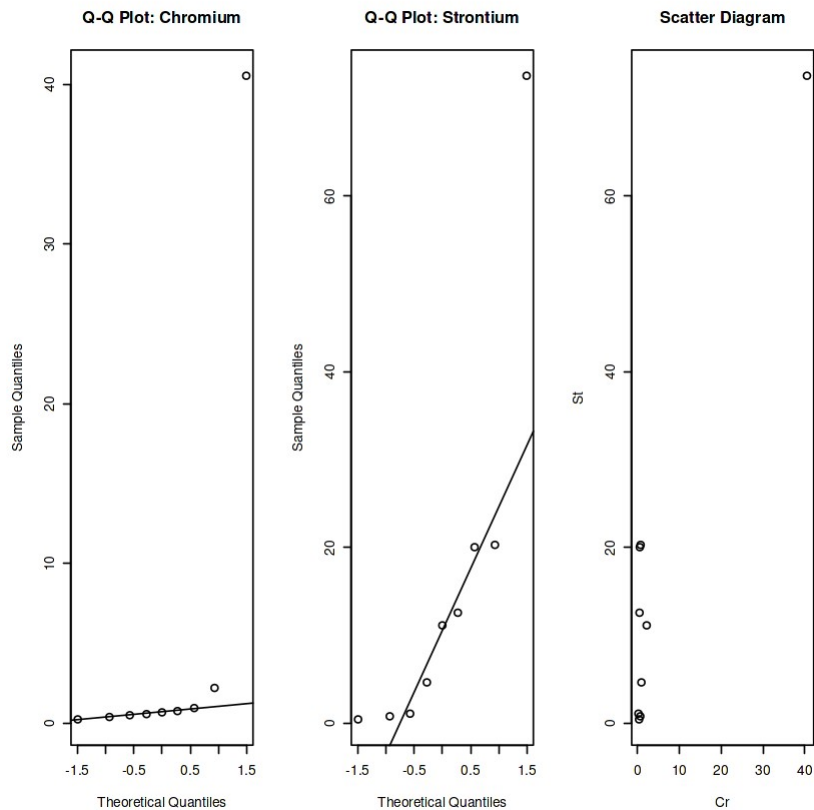
90% Joint Confidence Ellipse with Data Overlay



After Outlier Removal

T^2 statistic for [0.30, 10]: 5.307872

Critical Value: 8.081043



```
stiffness <- c(1232, 1115, 2205, 1897, 1932, 1612, 1598, 1804, 1752,
2067,
                2365, 1646, 1579, 1880, 1773, 1712, 1932, 1820, 1900,
2426,
                1558, 1470, 1858, 1587, 2208, 1487, 2206, 2332, 2540,
2322)
bending <- c(4175, 6652, 7612, 10914, 10850, 7627, 6954, 8365, 9469,
6410,
              10327, 7320, 8196, 9709, 10370, 7749, 6818, 9307, 6457,
10102,
              7414, 7556, 7833, 8309, 9559, 6255, 10723, 5430, 12090,
10072)
lumber <- data.frame(Stiffness = stiffness, Bending = bending)

n <- nrow(lumber)
p <- 2
alpha <- 0.05
x_bar <- colMeans(lumber)
S <- cov(lumber)

# a
alpha <- 0.05
F_crit <- qf(1 - alpha, df1 = p, df2 = n - p)
radius <- sqrt( ((n - 1) * p / (n - p)) * F_crit / n )
```

```
plot(lumber_data, xlab = "Stiffness", ylab = "Bending Strength",
     main = "95% Confidence Ellipse", pch = 16)
lines(ellipse::ellipse(S, centre = x_bar, t = radius), col = "blue",
      lwd = 2)
```

```
# b
```

```
mu_0 <- c(2000, 10000)
```

```
T2 <- n * t(x_bar - mu_0) %*% solve(S) %*% (x_bar - mu_0)
```

```
T2_crit <- ((n - 1) * p / (n - p)) * qf(1 - alpha, p, n - p)
```

```
cat("Test for Typical Values [2000, 10000]'\n")
```

```
cat("T^2 statistic:", as.numeric(T2), "\n")
```

```
cat("Critical Value:", T2_crit, "\n")
```

```
if(as.numeric(T2) > T2_crit) {
```

```
  cat("Result: Data is not consistent with these values (Reject H0)\n")
}
```

```
else {
```

```
  cat("Result: Data is consistent with these values (Do not Reject H0)\n")
}
```

```
# c
```

```
par(mfrow = c(1, 3))
```

```
qqnorm(lumber$Stiffness, main = "Q-Q Plot: Stiffness");
```

```
qqline(lumber$Stiffness)
```

```
qqnorm(lumber$Bending, main = "Q-Q Plot: Bending");
```

```
qqline(lumber$Bending)
```

```
plot(lumber$Stiffness, lumber$Bending, main = "Scatter Diagram",
     xlab = "Stiffness", ylab = "Bending Strength")
```

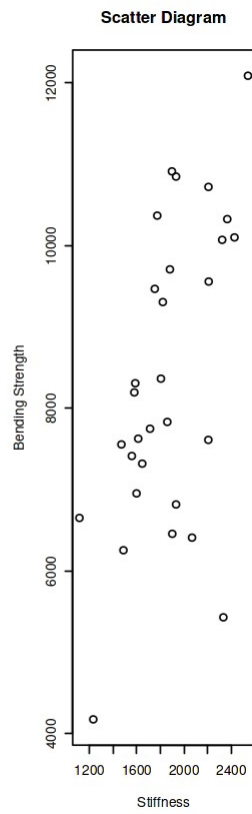
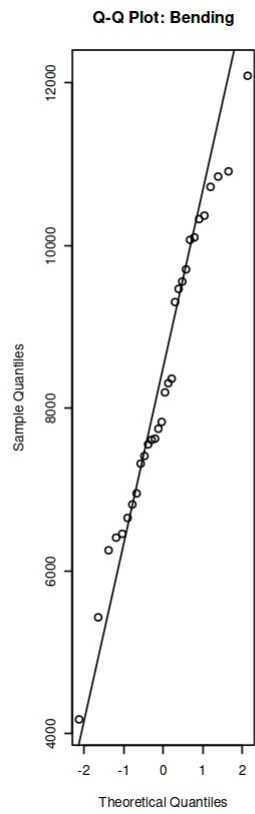
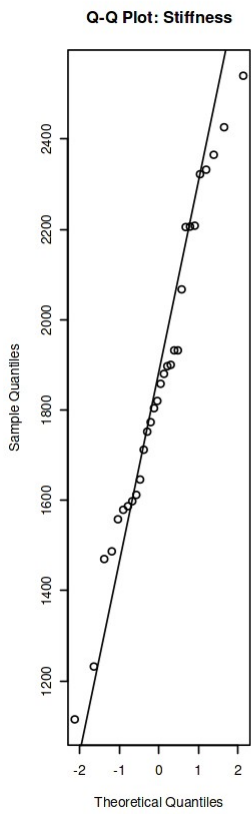
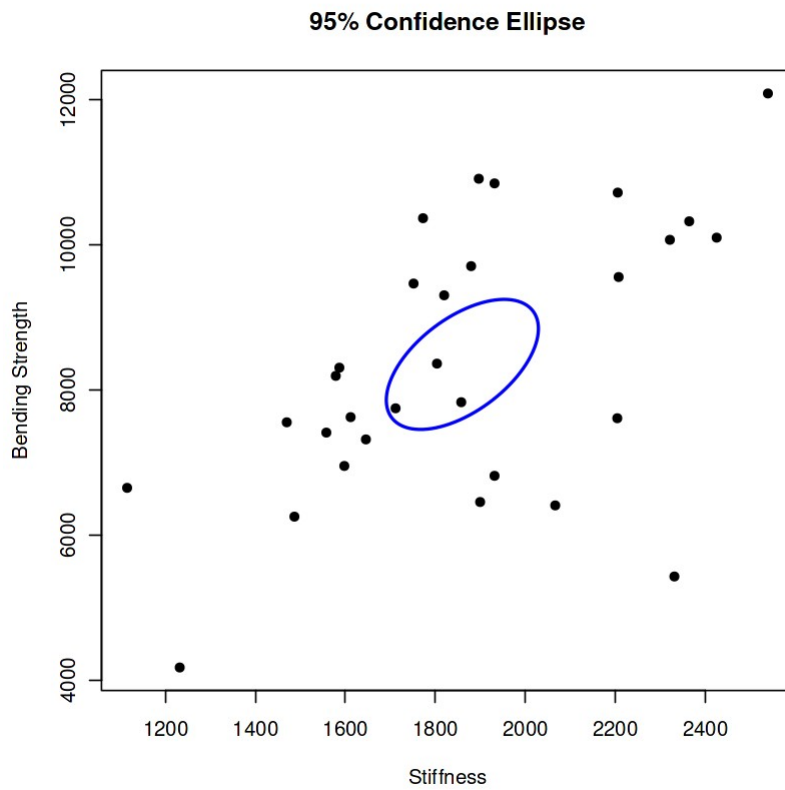
```
par(mfrow = c(1, 1))
```

```
Test for Typical Values [2000, 10000]'
```

```
T^2 statistic: 23.64778
```

```
Critical Value: 6.91937
```

```
Result: Data is not consistent with these values (Reject H0)
```



a. The 95% confidence ellipse was successfully created and plotted. It is centered around the sample mean and tilted upward, showing a positive relationship between lumber stiffness and bending strength.

b. The sample data does not match the proposed typical values of $\mu_{10}=2000$ for stiffness and $\mu_{20}=10,000$ for bending strength. The calculated Hotelling's T^2 value is 23.64778, which is higher than the critical value of 6.91937. Therefore, the null hypothesis is rejected. The point [2000, 10000]' would also be outside the 95% confidence ellipse.

c. The data appears to be reasonably consistent with a bivariate normal distribution. The Q-Q plots show that the values are close to the expected straight lines, while the scatter plot shows a roughly elliptical pattern with a positive relationship. There are also no major outliers, which supports the assumption of bivariate normality.